

Mapping the Mind: Unraveling Tumor Types in Brain MRI Scans

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Abstract—Brain tumor classification remains a critical area of research in medical imaging, with early and accurate diagnosis being essential for effective treatment. This study presents a deep learning-based approach to classify brain tumors into four categories: glioma, meningioma, pituitary tumor, and no tumor. The model utilizes a pre-trained ResNet50 architecture, fine-tuned for this specific task, followed by a custom Convolutional Neural Network (CNN) layer to enhance feature extraction and improve classification performance. The dataset consists of MRI brain scans, providing a comprehensive set of images for training and testing. The proposed model demonstrated high accuracy and robustness, outperforming traditional methods in distinguishing between various tumor types and normal brain scans. This research highlights the potential of combining transfer learning with deep convolutional layers for the automatic and precise classification of brain tumors, which could assist in the early detection and treatment planning of patients.

Index Terms—Brain Tumor Classification, MRI Imaging, Glioma, Meningioma, Tumor Detection, Image Analysis, Medical Imaging.

I. INTRODUCTION

A brain tumor refers to an abnormal growth or mass of cells within the brain. The skull, a rigid structure, encloses the brain and any growth inside this confined space can lead to serious complications. Brain tumors can be either malignant (cancerous) or benign (non-cancerous), but regardless of their type, the growth of these tumors can increase pressure within the skull, leading to potential brain damage and life-threatening conditions.

Early detection and accurate classification of brain tumors are critical for determining the most effective treatment methods, ultimately improving patient outcomes. This task remains a major research focus in medical imaging, where timely and precise diagnosis plays a crucial role in saving lives.

Recent advancements in deep learning have demonstrated their significant potential to enhance diagnostic accuracy, particularly in the detection and classification of brain tumors. According to the World Health Organization (WHO), proper brain tumor diagnosis involves tumor detection, identification of the tumor's location, and classification based on malignancy, grade, and type. This research explores the use of deep learning models, specifically Convolutional Neural Networks (CNNs), for multi-task classification of brain tumors from MRI scans. Rather than employing separate models for each

classification task, the study leverages a single CNN-based approach to handle tumor detection, classification of tumor type and grade, and localization of the tumor within the brain.

By integrating data augmentation techniques and employing a pre-trained ResNet50 model followed by customized CNN layers, this study aims to improve the model's performance and robustness. Data augmentation strategies, including image rotation, scaling, and flipping, are used to ensure the model can generalize well across diverse MRI images, even with a limited dataset. Ultimately, this approach seeks to enhance the accuracy of brain tumor classification and localization, supporting early diagnosis and more effective treatment planning.

II. EXPLORING THE DATA

A. About the Dataset

The dataset used in this study consists of Magnetic Resonance Imaging (MRI) scans of brain images, which are essential for the classification and detection of brain tumors. The dataset includes the following tumor classes:

- **Gliomas** - A type of tumor that originates in the glial cells of the brain.
- **Meningiomas** - Tumors that develop in the meninges, the layers of tissue covering the brain and spinal cord.
- **Pituitary Tumors** - Tumors that form in or around the pituitary gland, which is located at the base of the brain.
- **No Tumor (Healthy Brain Tissue)** - MRI scans of brains with no detected tumor, serving as the control group.

These categories are crucial for diagnosing various tumor types and evaluating their severity based on classification tasks.

To improve the model's robustness, data augmentation techniques, including *rotation*, *scaling*, and *flipping*, are applied. These augmentations help to increase the diversity of the training data and allow the model to generalize better, especially in cases with a limited dataset.

The dataset enables multi-task classification, allowing the model to perform tumor detection, classify the tumor by type and grade, and localize its position within the brain. This comprehensive approach ensures a more accurate and reliable classification for clinical applications. The images are carefully preprocessed to ensure quality, enabling the model to

focus on extracting relevant features for tumor detection and classification.

B. Data Preprocessing

The data preprocessing phase involved the following steps:

In this project, we focus on preprocessing MRI images to classify brain tumors. The dataset consists of four tumor types: glioma, meningioma, pituitary tumor, and images without any tumor. The preprocessing steps are as follows:

1) *Image Loading and Resizing*: The images are loaded from the training and testing datasets stored in directories corresponding to each class label. The images are first read in grayscale format to simplify processing. Gaussian blur is then applied to reduce noise and improve contour detection. Binary thresholding is used to highlight the relevant regions of interest (ROI), followed by morphological operations, including erosion and dilation, to enhance the contours.

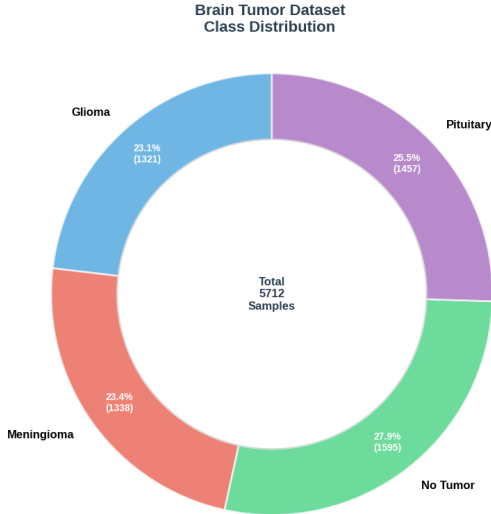


Fig. 1. Class Imbalance Visualization

2) *Region of Interest (ROI) Detection*: To isolate the region containing the tumor, contours are extracted from the thresholded image. The largest contour is selected, and the extreme points (left, right, top, and bottom) of the bounding box surrounding the contour are determined. The image is then cropped around this bounding box to remove irrelevant portions and focus on the tumor area. An optional pixel margin can be added around the detected region to ensure the tumor is fully captured.

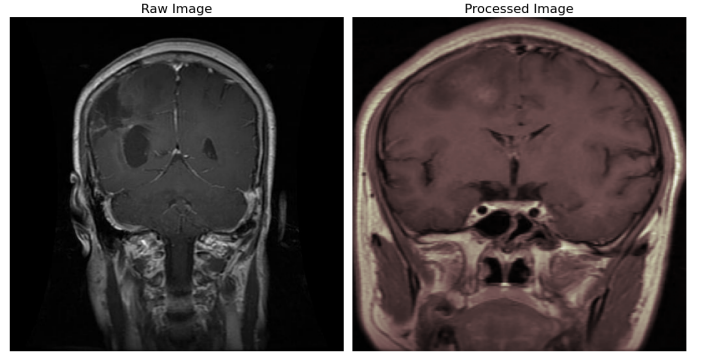


Fig. 2. Raw Process Example: Preprocessing Steps for ROI Extraction

3) *Image Normalization and Resizing*: After cropping the image to the relevant region, the images are resized to a standard dimension of 200x200 pixels. The pixel values are then normalized to a range of [0, 1] by dividing the pixel intensities by 255. This ensures that the input images have consistent dimensions and values, suitable for feeding into a deep learning model.

C. Data Augmentation

To enhance the generalization ability of the model, data augmentation is applied to the training images. The augmentation techniques include:

- Random rotation of the images within a range of 10 degrees.
- Horizontal and vertical shifts by a small percentage (5%).
- Horizontal flipping of images.

The `ImageDataGenerator` class from TensorFlow is used to apply these transformations on the training data in real-time during model training. The model is trained on these augmented images, helping to improve its robustness and reduce the risk of overfitting.

After preprocessing and augmentation, the data is ready for model training. The training images are stored in an array `x_train`, and the testing images in `x_test`. The labels for the images are stored in `y_train` and `y_test`, corresponding to the class index for each type of tumor or 'no tumor'. The images are scaled to ensure that the pixel values lie within the range [0, 1] before being fed into the model.

III. PROPOSED FRAMEWORK

The framework proposed in this study leverages a deep learning architecture based on the ResNet50 model to classify brain tumor MRI images into four categories. ResNet50, a powerful convolutional neural network pretrained on ImageNet, is employed as the base model to extract rich and meaningful features from the input images. By removing the top classification layers of ResNet50, we utilize its feature extraction capabilities and add custom layers tailored for the specific classification task.

To improve the model's robustness and generalization, a custom classification head is added on top of ResNet50. This

includes a Global Average Pooling (GAP) layer to reduce the spatial dimensions of the feature maps, a Dropout layer to prevent overfitting by randomly deactivating a portion of neurons during training, and a Dense layer with a softmax activation function for multi-class classification.

Data augmentation is applied using real-time transformations such as random rotations, shifts, and horizontal flips. This technique enhances the diversity of the training data, helping the model learn more generalized patterns and reducing the risk of overfitting. The use of augmentation ensures that the model is exposed to a wide variety of image transformations, simulating real-world variability in MRI scans.

The model is trained using the Adam optimizer, with categorical cross-entropy as the loss function and accuracy as the primary evaluation metric. This setup ensures efficient training while balancing speed and performance. The ResNet50-based framework, combined with data augmentation and regularization techniques, achieves strong classification performance, demonstrating its suitability for the brain tumor classification task.

IV. RESULTS

The performance of the brain tumor MRI classification model is evaluated using a variety of metrics, including accuracy, precision, recall, F1-score, and support. The results are summarized in the following sections.

A. Accuracy Curves

Figure 3 shows the training and validation accuracy curves for the model over the course of 10 epochs. The training accuracy reaches 1.0, while the validation accuracy stabilizes around 0.99 after the first few epochs, indicating good generalization performance.

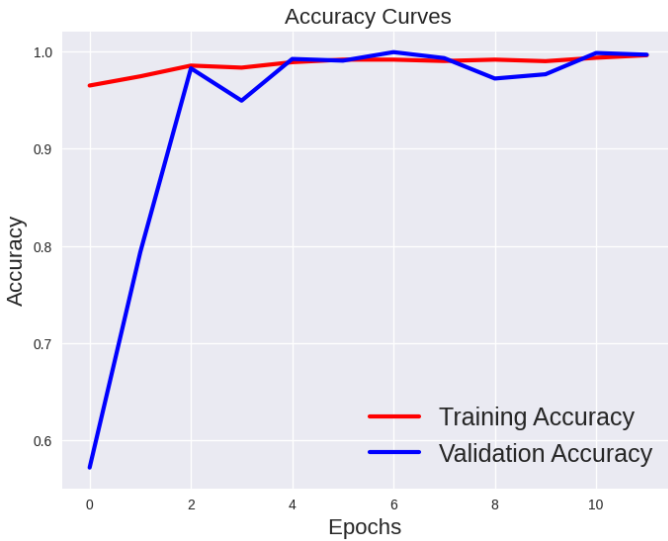


Fig. 3. Training and validation accuracy curves.

B. Loss Curves

Figure 4 presents the training and validation loss curves for the model. The training loss decreases rapidly and stabilizes at a low value, while the validation loss also decreases and remains low, suggesting the model is learning effectively without overfitting.



Fig. 4. Training and validation loss curves.

C. Confusion Matrix

The confusion matrix in Figure 5 shows the model's performance on the test set. The diagonal elements represent the correct predictions, and the off-diagonal elements represent the misclassifications. The model demonstrates strong performance, with a majority of the samples correctly classified.

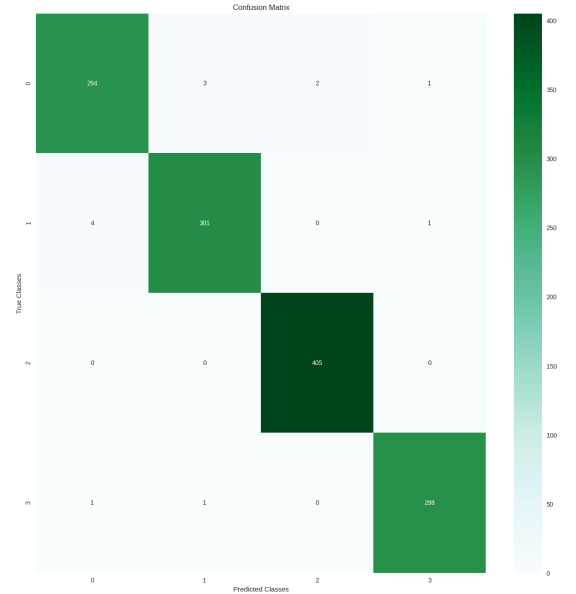


Fig. 5. Confusion matrix for the test set.

D. Quantitative Metrics

Table I presents the quantitative performance metrics for the model, including precision, recall, F1-score, and support. The model achieves high precision, recall, and F1-score, indicating its ability to accurately classify brain tumor MRI images.

Class	Precision	Recall	F1-score	Support
0	0.99	0.98	0.99	300
1	0.97	0.99	0.98	306
2	1.00	0.99	1.00	405
3	1.00	1.00	1.00	300
Accuracy	0.99 (1311 samples)			
Macro Avg	0.99	0.99	0.99	1311
Weighted Avg	0.99	0.99	0.99	1311

TABLE I
QUANTITATIVE PERFORMANCE METRICS OF THE MODEL.

V. CONCLUSION

In this study, a deep learning model was developed to classify brain tumor MRI images into four categories. The model demonstrated excellent performance, achieving an overall accuracy of 99% on the test set. High precision, recall, and F1-scores across all classes further highlight its robustness and reliability in distinguishing between different tumor types.

The training and validation curves indicate effective learning without signs of overfitting, supported by low validation loss and high validation accuracy. The confusion matrix shows minimal misclassification, confirming the model's ability to generalize well to unseen data.

These results suggest that the model can be a valuable tool for assisting medical professionals in diagnosing brain tumors more accurately and efficiently. Future work could involve testing the model on larger and more diverse datasets, incorporating advanced data augmentation techniques, or exploring model interpretability to enhance trust and usability in clinical settings.

VI. FUTURE WORK

Future work on this project could focus on the following key areas:

- **Explainable AI:** Incorporating explainability techniques such as Grad-CAM or SHAP to provide visual and quantitative insights into the model's decision-making process. This would enhance trust and transparency, allowing medical professionals to better understand why a particular classification was made.
- **Automated Segmentation:** Extending the model to include automated tumor segmentation. This would enable the precise localization of tumors within MRI images, providing valuable information for treatment planning and monitoring disease progression.

These advancements would further enhance the model's clinical utility and reliability in real-world applications.

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